

Calculus II

Integration by parts

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2019

Outline

1 Integration by Parts

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Integration by Parts

Every differentiation rule corresponds to a differential form rule which in turn corresponds to an integration rule.

$(uv)' = u'v + uv'$	Product Rule
$d(uv) = vdu + udv$	Differential Prod. Rule
$\int d(uv) = \int vdu + \int udv$	integration of the above
$uv = \int vdu + \int udv$	rearrange
$\int udv = uv - \int vdu$	

We just proved the following.

Proposition ((Rule of) Integration by Parts)

$$\int udv = uv - \int vdu$$

Integration by parts: strategy for applying

Integration by parts:

$$\int u dv = uv - \int v du.$$

Generally: Choose u in this order: **LIPET**

Logs, Inverse trig, Polynomial, Exponential, Trig

Integration by parts: $\int u dv = uv - \int v du.$

Example

$$\begin{aligned}\int x \sin x dx &= \int x d(-\cos x) && \left| \sin x dx = d(-\cos x) \right. \\ &= x(-\cos x) - \int (-\cos x) dx \\ &= -x \cos x + \int \cos x dx \\ &= -x \cos x + \sin x + C\end{aligned}$$

Integration by parts: $\int u dv = uv - \int v du.$

Example

$$\begin{aligned}\int \ln x dx &= (\ln x)x - \int x d(\ln x) && \left| \text{integrate by parts} \right. \\ &= x \ln x - \int x (\ln x)' dx \\ &= x \ln x - \int x \frac{1}{x} dx \\ &= x \ln x - \int dx \\ &= x \ln x - x + C.\end{aligned}$$

Integration by parts: $\int u dv = uv - \int v du.$

Example

$$\begin{aligned}\int t^2 e^t dt &= \int t^2 d(e^t) && \left| \text{integrate by parts} \right. \\ &= t^2 e^t - \int e^t d(t^2) \\ &= t^2 e^t - \int e^t 2t dt \\ &= t^2 e^t - 2 \int t d(e^t) && \left| \text{integrate by parts} \right. \\ &= t^2 e^t - 2 \left(t e^t - \int e^t dt \right) \\ &= t^2 e^t - 2 t e^t + 2 e^t + C\end{aligned}$$

Integration by parts: $\int u dv = uv - \int v du.$

Example

$$\begin{aligned}\int e^x \sin x dx &= \int \sin x d(e^x) \\&= (\sin x)e^x - \int e^x d(\sin x) \\&= e^x \sin x - \int e^x \cos x dx \\&= e^x \sin x - \int \cos x d(e^x) \\&= e^x \sin x - \left((\cos x)e^x - \int e^x d(\cos x) \right) \\&= e^x \sin x - \cos x e^x + \int e^x (-\sin x) dx \\&= e^x \sin x - \cos x e^x - \int e^x \sin x dx\end{aligned}$$

Integration by parts: $\int u dv = uv - \int v du.$

Example

$$\begin{aligned}\int e^x \sin x dx &= e^x \sin x - \cos x e^x - \int e^x \sin x dx \\ 2 \int e^x \sin x dx &= e^x \sin x - \cos x e^x \\ \int e^x \sin x dx &= \frac{1}{2} (e^x \sin x - \cos x e^x) + C\end{aligned}$$

Integration by parts: $\int u dv = uv - \int v du.$

Example

Set $w = 1 + x^2$.

$$\begin{aligned}
 \int_0^1 \arctan x dx &= [(\arctan x)x]_{x=0}^{x=1} - \int_0^1 x d(\arctan x) \\
 &= 1 \cdot \arctan 1 - 0 \cdot \arctan 0 - \int_{x=0}^{x=1} x \frac{1}{1+x^2} dx \\
 &= \frac{\pi}{4} - \int_{x=0}^{x=1} \frac{1}{1+x^2} d\left(\frac{x^2}{2}\right) \\
 &= \frac{\pi}{4} - \frac{1}{2} \int_{x=0}^{x=1} \frac{1}{1+x^2} d(1+x^2) \\
 &= \frac{\pi}{4} - \frac{1}{2} \int_{x=0}^{x=1} \frac{1}{w} dw = \frac{\pi}{4} - \frac{1}{2} [\ln |w|]_{x=0}^{x=1} \\
 &= \frac{\pi}{4} - \frac{1}{2} \left[\ln(1+x^2) \right]_{x=0}^{x=1} \\
 &= \frac{\pi}{4} - \frac{1}{2} (\ln 2 - \ln 1) = \frac{\pi}{4} - \frac{1}{2} \ln 2.
 \end{aligned}$$